

Detailed Lesson Plan • Session 1 • Sorting Algorithm Introduction

Course:	Sorting Algorithm Game	Session:	Session 1 of 1	Duration:	30 Minutes
Depth:	Beginner	Language:	English	Delivery:	Digital Game + Offline Worksheet

★ LEARNING OBJECTIVES

LO 1	RECALL	Define 'sorting' in simple words — e.g., 'putting things in order.'
LO 2	IDENTIFY	Recognise at least TWO sorting rules: Smallest→Biggest and Biggest→Smallest.
LO 3	APPLY	Arrange 3–5 everyday objects in the correct sorted order without teacher help.
LO 4	CONNECT	Relate sorting to ONE real-life Indian situation encountered daily (tiffin dabbas, assembly line, coins).
LO 5	ENGAGE	Complete at least one full game round independently, achieving a basic pass score of ≥ 60%.

◆ BLOOM'S TAXONOMY ALIGNMENT

L1: Remember	L2: Understand	L3: Apply	L4-6: Future
Name the word 'sorting' and its meaning	Explain WHY we sort things	Sort items on screen or with physical manipulatives	Reserved for Intermediate & Advanced sessions

■ KEY CONCEPTS WITH INDIAN EXAMPLES

CONCEPT 1 Concept 1 — What Is Sorting? • Sorting = putting things in a special ORDER. • Tiffin Dabbas: Amma stacks steel dabbas biggest at bottom, smallest on top — sorting by SIZE. • Assembly Line: Students stand shortest → tallest for PT Sir — sorting by HEIGHT. • Rupee Coins: Papa groups ₹1, ₹2, ₹5 coins separately — sorting by VALUE. • Sabzi Mandi: Vendor keeps big & small potatoes in separate piles — sorting by SIZE.	CONCEPT 2 Concept 2 — Sorting Rules (Algorithm) • RULE A — Ascending (Small → Big): rupee coins ₹1, ₹2, ₹5, ₹10, ₹20. • RULE B — Descending (Big → Small): stacking laddoos biggest first. • RULE C — Grouping (Same with Same): rangoli powder sorted by colour into cups. • Key Vocab: sort, order, smallest, biggest, shortest, tallest, first, last, middle. • NOTE: Do NOT use the word 'algorithm' in student-facing content.
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CONCEPT 3

Concept 3 — Why Do We Sort?

- Sorting SAVES TIME — Riya found her Panchatantra book in 5 seconds after sorting.
- Sorting AVOIDS CONFUSION — everything has its own place.
- Sorting MAKES LIFE EASIER — the core motivational message for learners.
- Story Hook: 'Riya's Messy Bookshelf' — problem → solution arc.

CONCEPT 4

Concept 4 — Steps of Sorting

- STEP 1: LOOK at all the things you want to sort.
- STEP 2: DECIDE your sorting rule (size? colour? height?).
- STEP 3: PLACE each item in the right position, one by one.
- Indian Parallel: Teacher calls Roll No. 1 first, then 2, then 3 — that is a RULE!

■ SESSION OUTLINE WITH TIMING (Total: 30 Minutes)

PHASE	TIME	ACTIVITY & CONTENT	FORMAT	FACILITATOR NOTES
Phase 1 Hook & Warm-Up	5 min	Animated story: 'Riya's Messy Bookshelf'. Question: 'Have you ever lost something because it was not in order?'	Animated Video (90 sec max)	Start with STORY not definitions. Hook question as speech bubble. Warm voice-over.
Phase 2 Concept Intro	7 min	Introduce sorting via tiffin dabba & assembly-line examples. Teach 6 key vocab words with visuals.	Guided Video + Illustrations	One concept at a time. Repeat key words ≥3 times. On-screen label for every new word.
Phase 3 Guided Practice	8 min	Game Round 1 (Teacher-led): Sort 3 tiffin dabbas by size. Hints ON — arrows point to correct slot.	Interactive Drag & Drop Game	Show 'Try it together!' banner. Celebrate partial success. Hints = directional arrows.
Phase 4 Independent Play	7 min	Game Round 2 (Student alone): Sort coins ■1, ■2, ■5, ■10, ■20 by value. Hints OFF.	Interactive Drag & Drop Game	NO countdown timer. 'Need Help?' button visible. Animated encouragement per correct placement.
Phase 5 Recap & Celebration	3 min	3-question quick quiz. Award 'Sorting Star ■' badge. Show 'Today I Learned' card for parents.	Animated Wrap-Up + Badge	Quiz = 1 MCQ + 2 drag-and-drop. Show stars ★★★ (NOT %). Badge awarded regardless of score.

■ INDIAN CONTEXT REFERENCES LIBRARY

Food & Kitchen	Steel tiffin dabbas (stacked by size), Idli sizes on a plate, Laddoos arranged smallest to biggest, Rotis sorted by thickness
School Life	Morning assembly height line, Roll number order in register, Pencils sorted by length, Notebooks stacked by size in bag
Market & Shops	Sabzi mandi vegetable sorting, Kirana shop same items together, Rupee coins by denomination (■1 ■2 ■5 ■10 ■20), Bangles by size
Festivals	Rangoli powder cups sorted by colour, Diyas arranged by size for Diwali, Flowers sorted for puja (big/small)
Character Names	Riya, Arjun, Meena, Chintu, Dadi, Amma, Papa, PT Sir, Sharmaji Uncle — reflect regional diversity
Regional Balance	Assets must represent North, South, East & West Indian students. Avoid defaulting to a single regional look.

✍ TONE AND STYLE GUIDE

■ Tone

- Primary: Warm, playful, encouraging, curious.
- Secondary: Clear, structured, confident — never bossy.
- Avoid: Academic, cold, test-like, intimidating.
- Feel like a friendly older sibling guiding through a game.

■ Language

- Max 8–10 words per instruction sentence.
- Always ACTIVE voice: 'Drag the small cup first.'
- Use WE/YOU: 'Let us sort together!'
- Repeat key vocab naturally; exclamations used sparingly.

■ Visuals

- Flat-design illustrations — NOT stock photography.
- Min 40% girl characters across the session.
- Rounded expressive faces; diverse Indian appearances.
- Primary colours dominant: red, blue, yellow, green.

■ Voice-Over

- Pace: 110–130 words per minute (slow & clear).
- Indian-accented English preferred and encouraged.
- Natural 2–3 second pauses after questions.
- Sound like a warm young adult teacher (25–35 yrs).

✗ THINGS TO AVOID (All Submissions Checked Against This List)

✗	Word 'Algorithm' in student content	Use 'sorting rule' or 'steps' instead. Technical term stays in educator notes ONLY.
✗	Countdown timers in the game	Causes anxiety. Use optional elapsed-time display only. Never a ticking countdown.
✗	Negative feedback sounds / red X marks	Use gentle redirection: 'Oops! Let us try again!' — no harsh buzzers or beeps.
✗	Western-only examples (pizza, baseball, Halloween)	All PRIMARY examples must be Indian. Western references only as optional secondary.
✗	Complex vocabulary in instructions	Max 2-syllable words preferred. New words always introduced with a visual or action.
✗	Gender stereotypes	'Mummy cooks / Daddy fixes' — both characters do ALL activities equally in Planrs content.
✗	Percentage scores shown to students	Use stars ★★ or smiley-face scales. Percentages are developmentally inappropriate.
✗	Long paragraphs on screen	Maximum 1 sentence at a time on screen for young learners.
✗	More than 5 items in Session 1 sorting tasks	Cognitive overload. Keep to 3–5 items maximum per round.
✗	Stock photography of real children	Use illustrated characters only — privacy, consent & brand consistency standards.
✗	Jump scares, loud unexpected sounds, flashing animations	Follow WCAG 2.1 Level AA guidelines for all digital assets. Sensory sensitivity applies.

■ MINIMUM EXPECTATIONS PER OUTPUT TYPE

6A. Hook Animation / Story Video

- Duration: 60–90 seconds • At least 1 named Indian child character (e.g., Riya)
- Setting: Recognisable Indian home or classroom environment
- Voice-over: Clear slow-paced English (max 100 words) • English subtitles mandatory
- Visual action: Objects must visually move from messy → sorted order on screen
- On-screen text: Max 5 words per frame • File format: MP4, minimum 720p resolution

6B. Concept Explanation Video / Slides

- Duration: 5–7 minutes • Minimum 2 Indian context examples from the reference library
- All 6 key vocabulary words appear on-screen WITH matching visuals (not text-only)
- Animated / illustrated visuals only — absolutely NO stock photography
- Minimum 1 pause-and-predict moment: 'What do you think happens next?'
- All images require alt-text descriptions • Reviewed by 1 child-development specialist

6C. Interactive Sorting Game (Digital)

- Minimum 2 rounds: Guided (hints ON) + Independent (hints OFF)
- Items per round: 3 in Round 1, 5 in Round 2 • Drag-and-drop (touch + mouse)
- Immediate feedback on each placement • Gentle correction — NO harsh sounds / red X
- Stars system (NOT %) at end of round • NO countdown timer whatsoever
- Loads within 5 seconds on 4G • Compatible: Android, iOS tablets + desktop browser

6D. End-of-Session Quick Quiz

- Exactly 3 questions: 1 Multiple Choice + 2 Drag-and-Drop
- Pass threshold: 60% (tracked internally — NOT shown to student)
- Explanation shown after EVERY question (correct or incorrect)
- Student may retry once • Sorting Star badge awarded on completion (not score-gated)
- Dashboard logs: score, time taken, retry count

6E. Printable Worksheet (Offline Support)

- 1 page, single-sided A4 • Cut-and-paste OR draw-and-arrange activity
- Uses familiar Indian objects: diyas, fruits, coins, tiffin dabbas
- Instructions in simple English + illustrated icons • Min font: 14pt body, 18pt heading
- Must be clearly usable when printed in BLACK & WHITE
- Short 2-line parent engagement note at bottom of page

6F. Teacher / Facilitator Guide (1-Page)

- Single A4 PDF, landscape preferred
- Must include: Objectives, Timing, Key Questions, Common Mistakes, Extension Ideas, Home Connect
- Written for a teacher with NO Computer Science background — zero jargon
- Include 1 tip for advanced learners + 1 tip for learners needing extra support
- Approved by 1 classroom teacher with early-years experience before production

✓ SUCCESS CRITERIA

A. Content Design Success Criteria (Pre-Launch)

- ✓ All 5 Learning Objectives addressable within the 30-minute session.
- ✓ At least 3 distinct Indian context examples appear in the session content.
- ✓ All 6 output types (6A–6F) meet every minimum specification listed in Section 7.
- ✓ Tone & Style Guide compliance checked across ALL assets before submission.
- ✓ All 11 'Things to Avoid' are absent from submitted content — zero exceptions.
- ✓ Reviewed by: 1 Curriculum Specialist + 1 Teacher + 1 Child Development Expert + 1 Accessibility Checker.
- ✓ A/B pilot test conducted with ≥5 learners before full platform release.

B. Student Learning Success Criteria (Post-Delivery)

- ✓ ≥ 80% of learners complete the full 30-minute session without exiting.
- ✓ ≥ 75% of learners achieve a pass score (≥ 60%) on the end-of-session quiz.
- ✓ ≥ 70% of learners can verbally explain 'sorting' when asked by a teacher post-session.
- ✓ Average Game Round 2 completion rate ≥ 85%.
- ✓ Average session satisfaction rating (emoji scale) ≥ 4 / 5.

C. Engagement & Product Success Criteria

- ✓ Session drop-off rate < 20% (started but did not finish).
- ✓ 'Sorting Star' badge claimed by ≥ 90% of learners who complete the session.
- ✓ Parent dashboard: 'Today I Learned' card shared by ≥ 50% of learners.
- ✓ Teacher feedback survey score ≥ 4.0 / 5.0 for 'Curriculum Alignment'.
- ✓ Zero accessibility complaints in first 2 weeks post-launch.
- ✓ Content flagged for revision if < 60% of learners pass the quiz (triggers review cycle).

✓ PRODUCTION CHECKLIST & DOCUMENT CONTROL

CURRICULUM

- Learning Objectives written & Bloom's mapped
- All 4 Key Concepts documented with Indian examples
- Session Outline finalised with 5-phase timing
- Indian Context References Library reviewed

CONTENT ASSETS

- Hook Animation script written & reviewed
- Concept Video storyboard approved
- Game Design Document (GDD) submitted
- Quiz questions (3 Qs) reviewed for age-appropriateness
- Worksheet designed & B&W; print tested
- Teacher Guide 1-pager drafted & peer-reviewed

QUALITY GATES

- Tone & Style Guide compliance checked
- All 'Things to Avoid' final check completed
- Accessibility review done (game + video + WS)
- Regional diversity audit passed
- Pilot test with ≥5 learners completed
- Curriculum Lead sign-off obtained

Document Control

VERSION	DATE	AUTHOR	CHANGES	STATUS
1.0	May 2026	Curriculum Design Team, Planrs Academy	Initial Draft — Released for Production	APPROVED

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